

Diagram illustrating the Tucker decomposition of a 3D tensor T into a core tensor G and three factor matrices $A_1, A_2, \dots, A_r$.

The tensor T is shown as a purple cube, labeled T , with dimensions $\in \mathbb{R}^{n \times m \times p}$.

The decomposition is represented by the equation:

$$T = A_1 \cdot G \cdot A_2 \cdot \dots \cdot A_r$$

where:

- A_1 is a blue vertical bar representing a factor matrix.
- A_2 is a blue vertical bar representing a factor matrix.
- $\dots$ represents the continuation of the factor matrices.
- A_r is a blue vertical bar representing a factor matrix.
- G is a green cube representing the core tensor.
- c_1 is a green diagonal bar representing the core tensor's dimensions along the first mode.
- b_1 is an orange horizontal bar representing the core tensor's dimensions along the second mode.
- c_2 is a green diagonal bar representing the core tensor's dimensions along the first mode.
- b_2 is an orange horizontal bar representing the core tensor's dimensions along the second mode.
- $\dots$ represents the continuation of the core tensor's dimensions.
- c_r is a green diagonal bar representing the core tensor's dimensions along the first mode.
- b_r is an orange horizontal bar representing the core tensor's dimensions along the second mode.